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Project Report

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INTRODUCTION

The aim of this report is to investigate a dataset of 105 anonymised Exchange-Traded Funds (ETFs) through the lens of their risk-adjusted performance, as well as to uncover their composition with respect to main asset classes, spanning broad equity indices, fixed-income instruments, and key commodities. The report highlights the relative efficiency of ETFs by considering both their return potential and risk profiles. By comparing top-performing ETFs with bottom performers, the analysis sheds light on the key characteristics that differentiate them, with particular emphasis on the role of asset class exposure, volatility, and risk management strategies in shaping ETF performance. The analysis is conducted using the Python programming language, leveraging tools from statistical inference, statistical learning, and optimisation.

The report is structured into four sections. **Section 1** presents the raw ETF and asset class data to identify the global dynamics of financial markets over the investigated period and to address potential data quality issues. It also provides a brief overview of the nature of the assets included. **Section 2** focuses on the calculation of key performance metrics, including log returns, volatility, maximum drawdown, and Sharpe and Sortino ratios. The revealed risk–reward trade-offs are discussed and ETFs are classified into best and worst performers based on these metrics. **Section 3** applies correlation, regression, and Principal Component Analysis (PCA) to explore the drivers of ETF performance by examining their relationships with the main asset classes. Finally, **Section 4** identifies two mystery portfolio allocations among the ETFs using both static and dynamic (rolling-window) quadratic optimisation. The resulting allocations are evaluated using goodness-of-fit metrics and are further analysed using the asset identification framework developed in Section 3.

1. Exploratory data analysis

The section examines key properties of the data used in the analysis and provides basic insights about the instruments included. The analysis is based on three datasets:

- ETFs - daily price data for 105 anonymised ETFs, normalised to 100 at the start;
- Main asset classes - daily price data for 14 major asset classes, including equity indices, fixed income instruments and commodities;
- Mystery allocations - daily price data for two portfolios allocated across ETFs,

The ETF dataset spans the period from 1 January 2019 to 20 May 2024, whereas the main asset dataset covers 1 January 2018 to 22 May 2024. To ensure consistency and comparability across analyses, both datasets were aligned to the common ETF date range, resulting in a unified observation window. No missing observations were identified.

Figure 1 presents the standardised price series (z-scores) of the anonymised ETFs over the sample period. The vast majority of instruments exhibit broadly similar stochastic dynamics, characterised by non-stationary price paths following shared stochastic trend. This behaviour is consistent with that of actively traded financial assets, with overall sentiment similarly influencing the performance of a wide range of assets.

Standardisation reveals the presence of four ETFs exhibiting markedly atypical behaviour. Unlike the rest of the universe, these instruments display near-linear trajectories with pronounced kinks and minimal variability over the entire five-year horizon. A closer inspection of their raw price and return series confirms that these ETFs remain almost unchanged throughout the sample period, exhibiting negligible volatility and marginal cumulative returns. Such characteristics suggest that these instruments are either illiquid, structurally constrained, or represent cash-like or synthetic products that do not reflect meaningful market exposure. As their behaviour is fundamentally inconsistent with the remainder of the ETF universe and may distort subsequent risk, correlation, and factor analyses, these four ETFs are excluded from all further work.

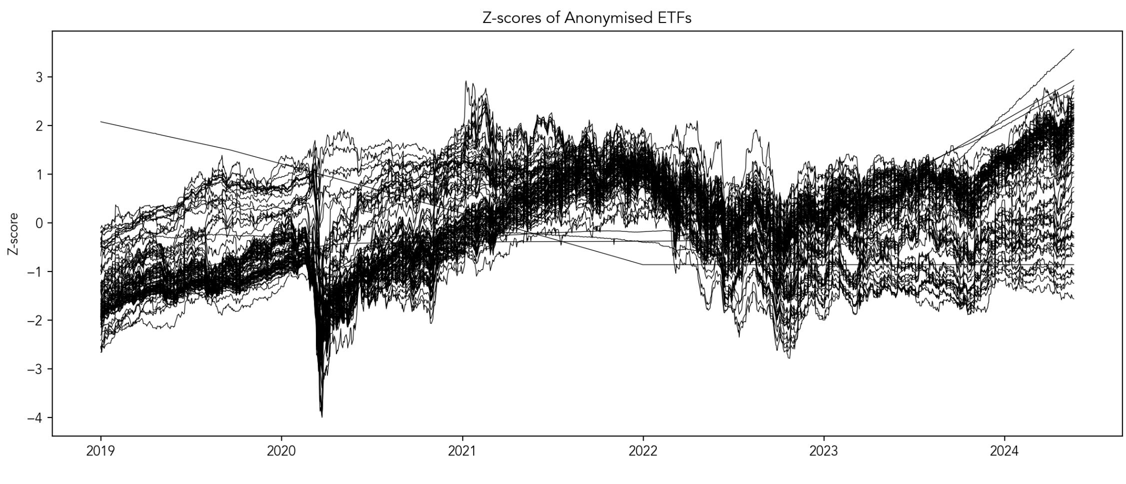

Figure 1. Z-scores for ETF daily price data.

Source: Own elaboration

Figure 2 illustrates the price dynamics of the main asset classes, while Figure 3 summarises their key risk and performance metrics. Despite differences in asset type—ranging from risk-on equities to traditionally defensive bonds and commodities—the assets display broadly similar trend fluctuations over the sample period. The performance metrics highlight systematic differences across asset classes, with equities exhibiting both higher annualised returns and higher volatility than global bonds. Among equities, the S&P 500 and the technology-oriented Nasdaq 100 emerge as the strongest performers, while European bonds deliver the weakest returns alongside the lowest volatility. The highest risk levels are observed for US small-capitalisation equities and the Nasdaq.

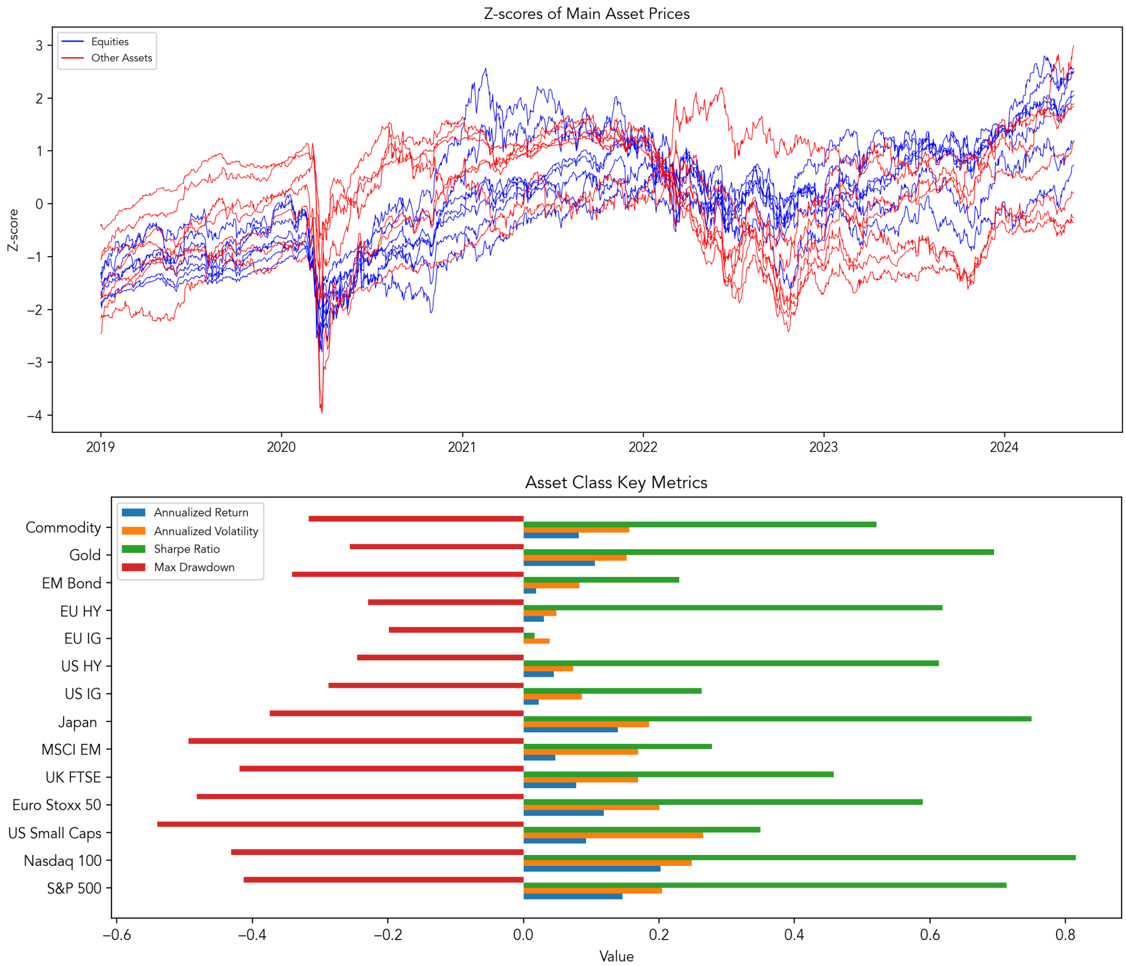

Figure 2. Z-scores for daily asset prices data with key metrics.

Source: Own elaboration

2. Performance and risk metrics

2.1. Variable calculation

The analysis is conducted using daily logarithmic returns, given by:

$$r_{t,i} = \ln(\frac{P_{t,i}}{P_{t,i-1}})$$

where $r_{t,i}$ - daily log return on i-th ETF between dates t-1 and t, computed from daily price data to ensure time-additivity and facilitate aggregation across different horizons. Log returns are particularly suitable for financial time-series analysis, as they stabilise variance and allow cumulative performance to be expressed as simple sums over time. From there we calculate ETFs' volatility by:

$$\sigma_i = \sqrt{\mathrm{Var}(r_{t,i})}$$

where σ_i - is a standard deviation of daily log returns of i-th ETF. For annualised metrics we get:

$$r_{i,ann} = \bar{r}_i \cdot 252 \qquad \sigma_{i,ann} = \sigma_i \cdot \sqrt{252}$$

where $\bar{r}_i$ is a mean daily log return of i-th ETF. We assume 252 trading days per year, in line with standard market conventions for daily financial data. Finally, we calculate two risk-reward tradeoff measures — Sharpe and Sortino ratios.

The Sharpe ratio measures the excess return per unit of total risk, capturing how efficiently an asset compensates investors for overall volatility. It treats upside and downside fluctuations symmetrically, making it most appropriate for broadly diversified portfolios. It is calculated by dividing annualised returns by annualised volatility, assuming a risk-free rate at $r_f = 0$:

$$\text{Sharpe}_i = \frac{r_{i,ann} - r_f}{\sigma_{ann}}$$

The Sortino ratio refines the Sharpe ratio by considering only downside risk (i.e. standard deviation of negative log returns: σ_i^-), penalising negative deviations from the target return while ignoring upside volatility. Like the Sharpe ratio, it is computed as the ratio of annualised returns to annualised downside deviation given zero risk-free rate, providing a more focused measure of downside-adjusted performance.

$$\text{Sortino}_i = \frac{r_{i,ann} - r_f}{\sigma_i^-}$$

2.2. Returns

Figure 3 displays cumulative sum of log returns, followed by a histogram of total performance of ETFs. Despite high correlation between ETF paths (following on average same market fluctuation like COVID-19 driven sell-off and gradual recovery).

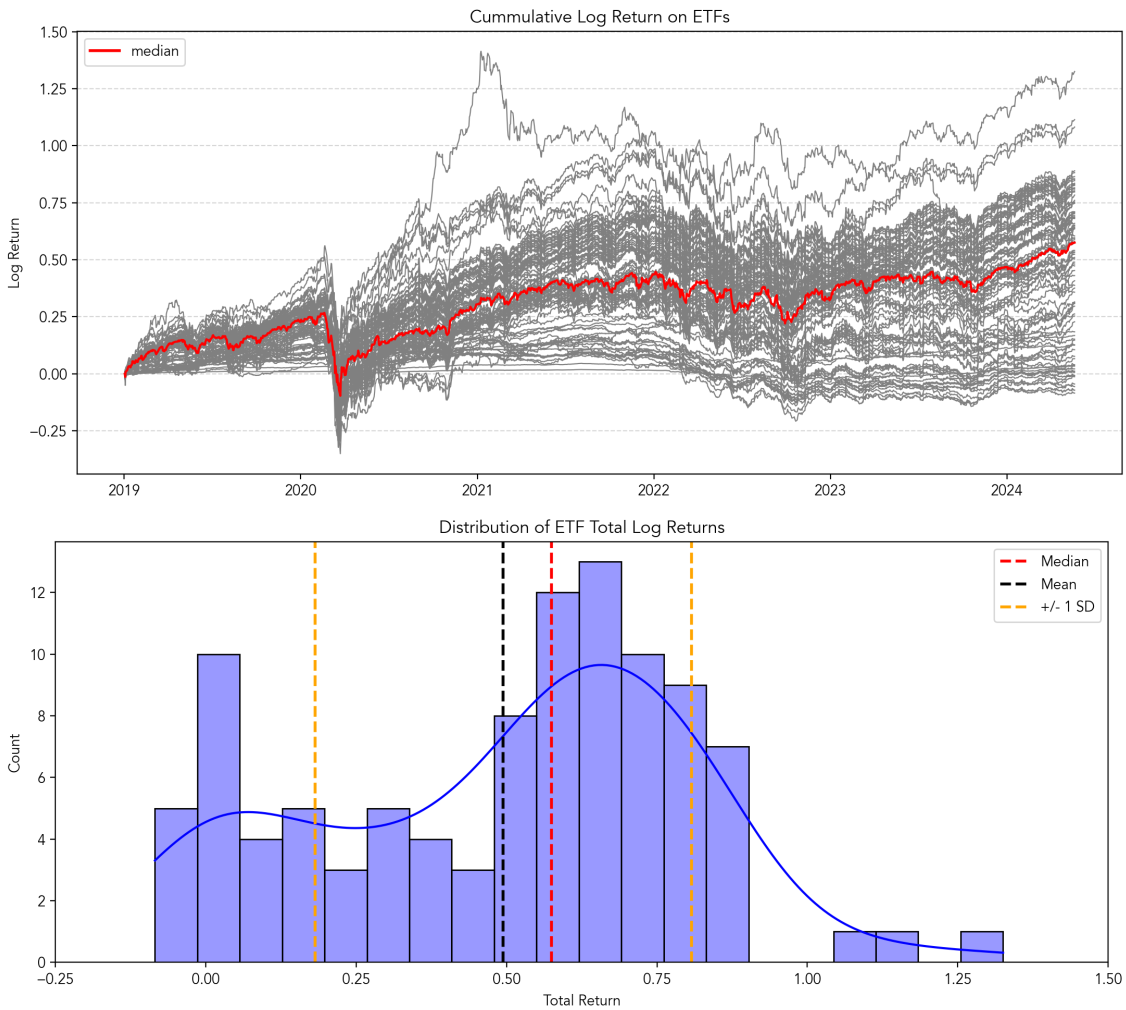

Figure 3. ETF cumulative log returns with distribution of total of returns.

Source: Own elaboration

In real terms, over the analysis period, ETF 53 was the star performer, achieving a total gain of approximately 276.3%, given by: $(e^{TotalReturn} - 1) \cdot 100\,\%$). ETF 12 and ETF 74 also posted stellar results, with total gains of about 204.7% and 195.0%, respectively. On the negative side, eight ETFs (indexed: 20, 39, 45, 70, 86, 90, 91, 103) posted losses over analysed period, with the worst one trading 8,13% below its initial value. The median performer achieved 77,71% return.

Table 1. Descriptive statistics for total log returns

N	Mean	Std	Min	25%	50%	75%	Max
101	0.494	0.313	−0.085	0.253	0.575	0.709	1.325

Source: Own elaboration

Interestingly, the histogram of total returns exhibits a bimodal pattern, with weaker-performing ETFs clustering roughly one standard deviation below the mean, while the stronger-performing assets lie slightly above the sample median. This separation suggests the presence of distinct risk/return profiles within the ETF universe, which may be driven by differences in volatility, asset class exposure, or strategy — factors that will be explored in more detail in subsequent sections.

We also examined the risk-adjusted performance of ETFs by dividing cumulative log returns by annualised volatility (Figure 4). This approach allows for a direct comparison of return efficiency, showing how much return each ETF delivers per unit of risk taken. It highlights ETFs that achieve strong performance without simply taking on the highest levels of risk. The histogram of risk-adjusted returns is much smoother and right-skewed, indicating that most ETFs deliver moderate efficiency, while a few stand out with exceptionally high risk-adjusted performance.

In this risk-adjusted framework, ETF 15 leads the ranking with a score of 5.16, outperforming even the overall top performer, ETF 53, which now ranks second with 4.93. Worth noting that ETF 15 yielded only 13.36% over the entire period. Some previously high-ranking ETFs experienced notable shifts: ETF 12, previously second, has fallen to fifth, while ETF 74, formerly third, has dropped out of the top five to eighth place. Meanwhile, new entrants have moved up in the risk-adjusted rankings: ETF 84 rises to third (from tenth) and ETF 97 climbs to fourth (from nineteenth).

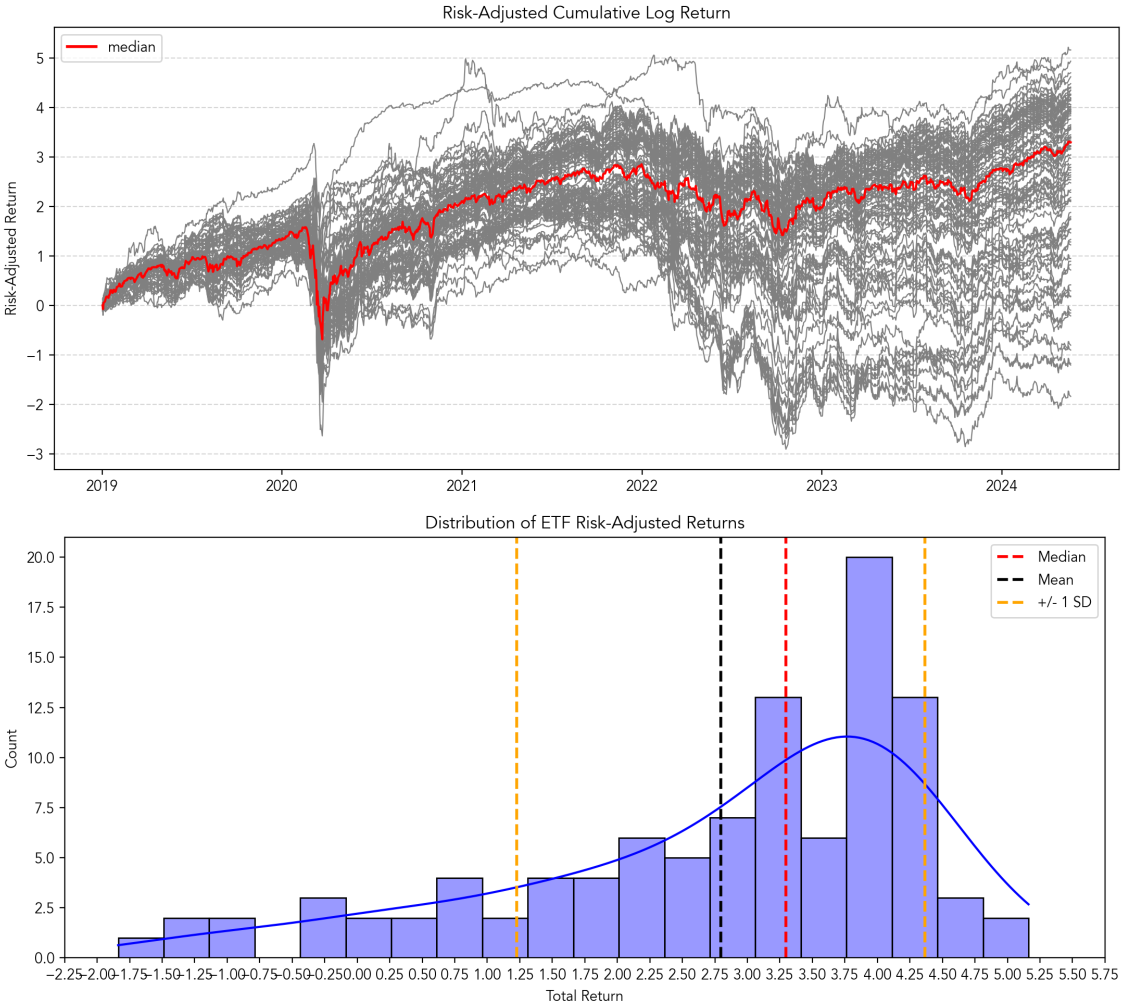

Figure 4. ETF risk-adjusted cumulative log returns with distribution of total of returns.

Source: Own elaboration

Table 2. Summary of top 5 performing ETFs

Rank	Total Return			Risk-Adjusted Return		
	ETF	Value	Total % Gain	ETF	Value	Total % Gain
1	ETF 53	1.33	276.22	ETF 15	5.16	13.36
2	ETF 12	1.11	204.65	ETF 53	4.93	276.38
3	ETF 74	1.08	195.06	ETF 84	4.70	129.68
4	ETF 23	0.89	143.51	ETF 97	4.62	113.90
5	ETF 58	0.88	141.57	ETF 12	4.48	204.72

Source: Own elaboration

2.3. Risk-reward tradeoff

The analysis of annualised metrics reveals a complex, yet intuitive relationship between risk and returns (Figure 5). The histograms show both annualised metrics exhibits a distinct bimodal distribution, even more pronounced for the volatility, with a clear cut-off between low-risk (below 0.10) and high risk clusters (between 0.15 and 0.25)

This difference in distribution is visualised in the Risk-Return scatter plot, which highlights two distinct clusters: a low-risk, low-return segment where volatility is constrained below 0.10, and a high-risk segment where volatility ranges from 0.15 to 0.25. The top 3 performing ETFs are all concentrated in the highest-volatility clusterIn this high-risk segment, but the 54% correlation between returns and volatility flags a limited ability to achieve higher returns by taking on additional volatility. If we consider only the medium cluster without top 3 ETFs, the risk-rewards correlation drops to 37%, indicating that in many case additional volatility remains unrewarded.

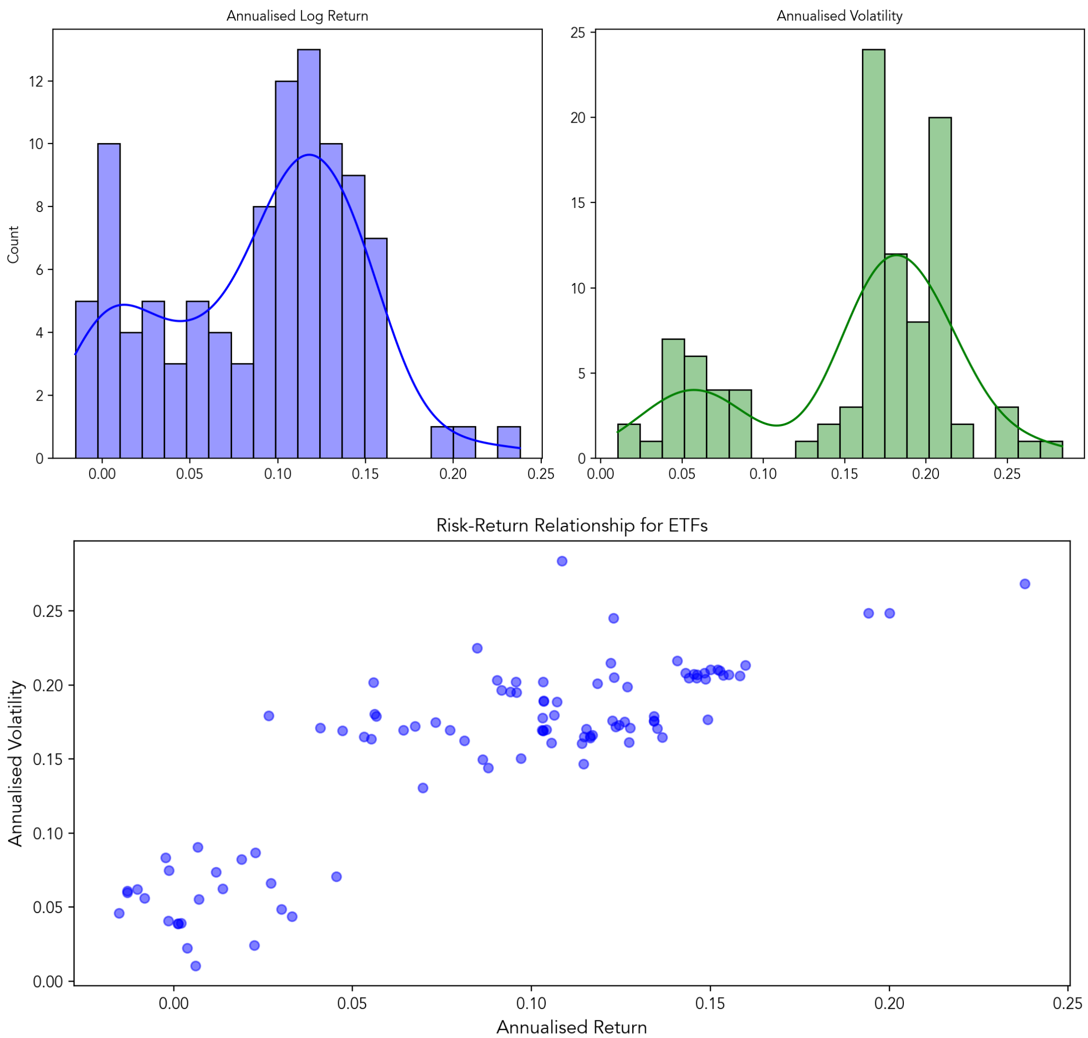

Figure 5. Annualised metrics: distribution and relationship.

Source: Own elaboration

Figure 6 relates the annualised metrics to maximum drawdown, a critical measure of downside risk. The left plot shows only a very weak positive correlation between annualised log return and max drawdown, meaning that higher average returns did not reliably equate to better downside protection. Returns largely clustered between 0.05 and 0.15 regardless of the drawdown experienced. Conversely, the right scatter plot demonstrates a strong, positive linear correlation between annualised volatility and max drawdown. ETFs with higher overall volatility almost universally experienced greater peak-to-trough losses. Volatility serves as a reliable proxy for downside risk in this ETF universe, as the safest ETFs (volatility below 0.10) maintained drawdowns below 0.20, while the most volatile funds incurred drawdowns ranging from 0.40 to over 0.90.

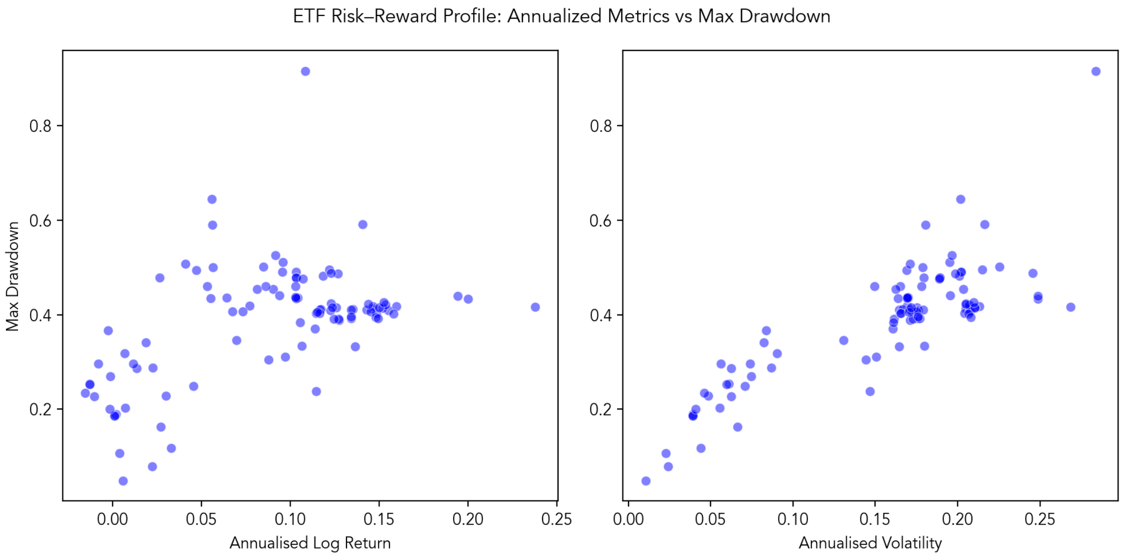

Figure 6. Max drawdown compared to annualised metrics.

Source: Own elaboration

2.4. Sharpe and Sortino ratios

The analysis of risk-adjusted returns highlights a critical divergence between total risk and downside risk (Figure 7). The histograms show that while the Sharpe Ratio is slightly skewed towards higher values (median around 0.6), the Sortino Ratio exhibits a distinct bimodal distribution with a higher central tendency (median around 1.4), distinguishing clearly between lower-efficiency funds and a larger cluster of high-efficiency performers.

The scatter plot further clarifies this relationship by comparing each ETF against the 45° diagonal. Methodologically, ETFs lying close to this line would exhibit similar Sharpe and Sortino ratios, indicating downside risk comparable to total volatility. However, the vast majority of ETFs in this universe are located significantly above the diagonal, where the Sortino Ratio exceeds the Sharpe Ratio. This suggests that for these funds, a large portion of their total volatility stems from positive upside variance rather than downside risk. Conversely, only a few ETFs fall below the diagonal—a zone that would indicate pronounced negative return episodes—confirming that the observed volatility was largely "good" volatility associated with upward price movements.

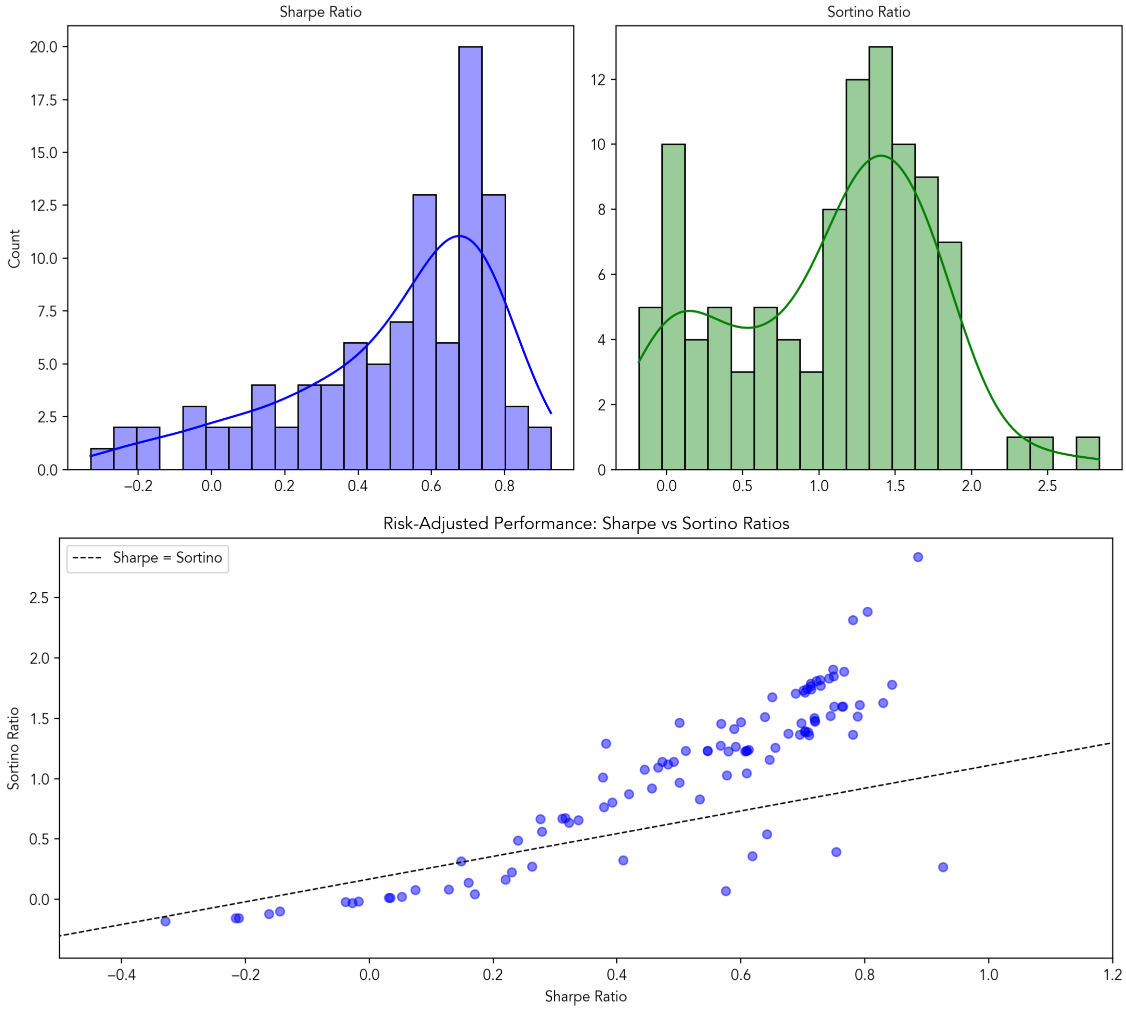

Figure 7. Sharpe vs Sortino: distribution and relationship.

Source: Own elaboration

2.5. Summary classification

Table 3 contains the top and bottom 5 ETFs for selected risk and performance metrics. The ranking highlights a clear distinction between aggressive growth drivers and highly efficient, low-volatility assets. ETF 53 and ETF 12 dominate the growth category, ranking as the top two funds for Cumulative Log Return and the Sortino Ratio. However, they also appear in the top five for Annualised Volatility, indicating their superior performance required accepting significant total risk.

In contrast, ETF 15 achieves the highest Sharpe Ratio (0.93) without chasing high raw returns, but rather by maintaining one of the lowest annualised volatilities (0.02) in the entire universe, effectively identifying it as the most efficient "safe haven" asset. Conversely, the bottom of the table is remarkably consistent: ETF 91, ETF 70, and ETF 39 rank as the worst performers across nearly every metric, offering negative nominal returns alongside the lowest risk-adjusted efficiency.

Table 3. Summary classification of ETFs by chosen metrics

Metric	Top 5	Bottom 5
Cumulative Log Return *(Total Growth)*	**ETF 53** (1.33) **ETF 12** (1.11) **ETF 74** (1.08) **ETF 23** (0.89) **ETF 58** (0.88)	**ETF 91** (-0.08) **ETF 70** (-0.07) **ETF 39** (-0.07) **ETF 90** (-0.06) **ETF 20** (-0.05)
Annualized Volatility *(Highest vs. Lowest Risk)*	**ETF 13** (0.28) **ETF 53** (0.27) **ETF 74** (0.25) **ETF 12** (0.25) **ETF 75** (0.25)	**ETF 65** (0.01) **ETF 43** (0.02) **ETF 15** (0.02) **ETF 78** (0.04) **ETF 57** (0.04)
Sharpe Ratio *(Risk-Adjusted Return)*	**ETF 15** (0.93) **ETF 53** (0.89) **ETF 84** (0.84) **ETF 97** (0.83) **ETF 12** (0.80	**ETF 91** (-0.33) **ETF 70** (-0.22) **ETF 39** (-0.21) **ETF 90** (-0.16) **ETF 20** (-0.14)
Sortino Ratio *(Downside Efficiency)*	**ETF 53** (2.84) **ETF 12** (2.38) **ETF 74** (2.31) **ETF 23** (1.90) **ETF 58** (1.89)	**ETF 91** (-0.18) **ETF 70** (-0.15) **ETF 39** (-0.15) **ETF 90** (-0.12) **ETF 20** (-0.10)

Source: Own elaboration

3. Relationship analysis

3.1. Correlation of ETFs with main asset classes

The correlation heatmap between the 100 ETFs and major global asset classes reveals a highly polarised investment universe. The upper section of the map is characterised by Equity Dominance, where a large cluster of "Pure Equity" funds exhibits strong positive correlations (dark red) with Developed Market indices (S&P 500, Nasdaq 100, Euro Stoxx 50) and High Yield credit, confirming that a significant portion of the sample is driven by aggressive market beta. Conversely, the bottom section defines a clear Fixed Income / Low-Risk path, where ETFs show high correlations with Investment Grade bonds (US IG, EU IG) and sovereign debt, while often displaying negative or low correlations with equities. Sandwiched between these two distinct blocks is vague middle strip, representing broadly diversified or multi-asset portfolios. In this transitional zone, correlations are moderate (lighter orange/red) across both asset types, indicating that true "all-weather" or balanced strategies are less prevalent in this dataset than pure directional bets.

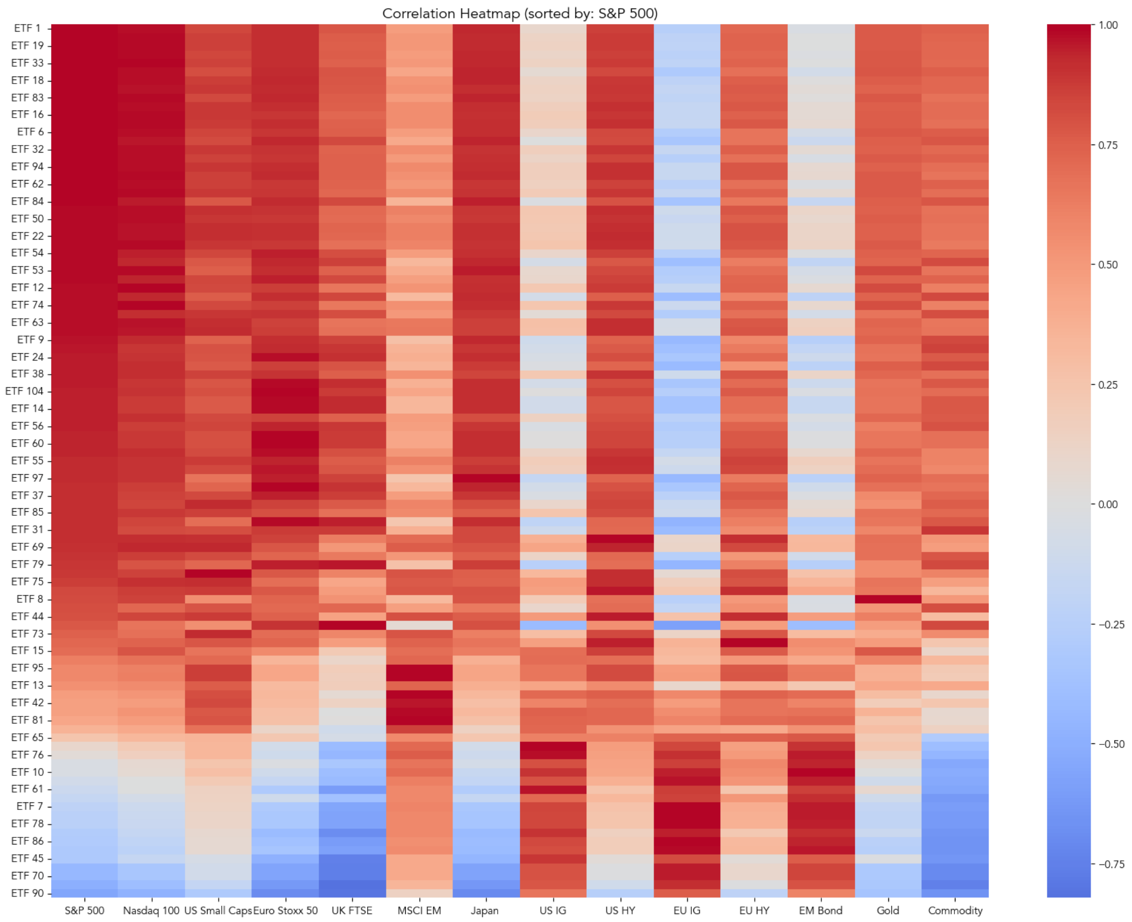

Figure 8. Heatmap with correlations between ETFs and main asset classes.

Source: Own elaboration

3.2. Linear regression based contributions

To extend the analysis beyond simple point-wise correlations, we estimated a Multiple Linear Regression (MLR) model for each ETF to isolate the marginal contribution of each asset class to performance. The model equation is as follows:

$$r_{t,i} = \alpha + \Gamma \mathbf{X}_t + \epsilon_t$$

where daily log return of the i-th ETF at time t is regressed on the matrix $\mathbf{X}$, comprising the daily log returns of the main asset classes. The use of log returns ensures stationarity and constrained variance, thereby satisfying the core stability assumptions required by the Gauss-Markov theorem for the Ordinary Least Squares estimator.

The resulting matrix of statistically significant regression coefficients confirms and sharpens the polarised market structure identified previously. The upper section of the universe exhibits a distinct Equity Beta profile, characterised by large, positive, and statistically significant coefficients for the S&P 500, Nasdaq 100, and Euro Stoxx 50, indicating that these funds are essentially leveraged plays on developed equity markets. Conversely, the lower section reveals a clear Credit/Duration cluster, where equity betas vanish and are replaced by significant positive loadings on US and EU Investment Grade and High Yield bond, as well as more geographically diversified equities. This clear separation in significant coefficients validates that the majority of these ETFs are driven by distinct, non-overlapping primary risk factors—either global equity risk or credit spread/interest rate risk—with little evidence of complex, multi-asset alpha generation in the middle of the distribution.

The decomposition of returns via OLS contributions (Figure 9) provides a critical link to our earlier risk-reward analysis, revealing clear segmentation in performance drivers. Top-tier performance was predominantly fueled by US equities, with specific tilts toward high-beta growth and small-cap exposures, while the middle tier relied on diversified global value stocks and the lower tail was dominated by bond-heavy portfolios. However, these results must be interpreted with caution due to the significant multicollinearity among asset classes (e.g., S&P 500 and Nasdaq 100), which can distort the signs of individual coefficients. Despite this statistical limitation, the model robustly confirms the fundamental trade-off: superior nominal returns in this universe were inextricably linked to accepting the higher volatility inherent to equity markets compared to fixed income.

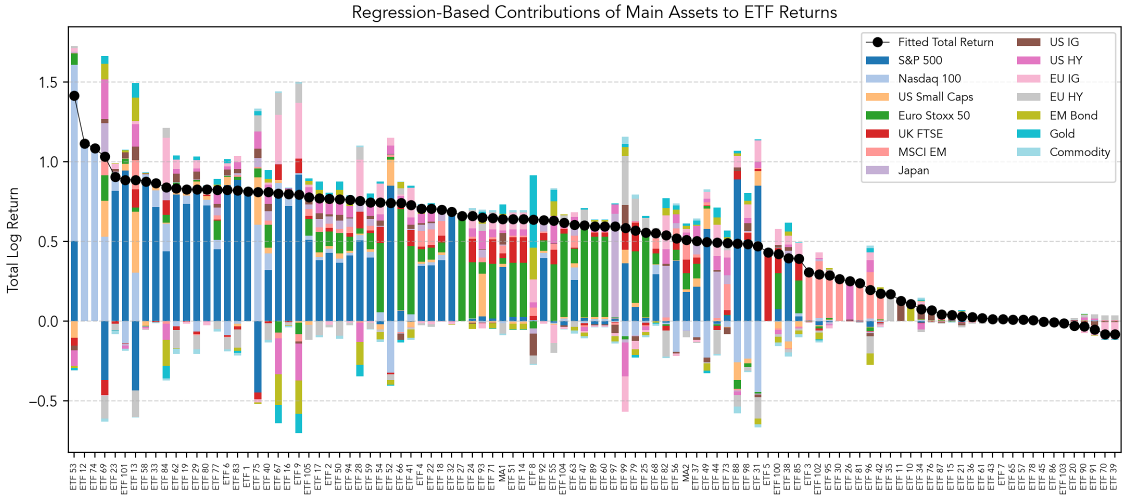

Figure 9. OLS-based contributions to ETF performance.

Source: Own elaboration

3.3. Principal Component Analysis

3.3.1. Methodology

To resolve the multicollinearity issue inherent in the OLS framework and identify orthogonal drivers of risk and return, we employ Principal Component Analysis (PCA). PCA is a dimensionality reduction technique that transforms the original set of correlated variables (asset class returns) into a new set of uncorrelated variables called principal components.

Mathematically, let X be the T x N matrix of standardised daily log returns for the $N = 14$ asset classes over T time periods. The first step involves computing the covariance matrix (or correlation matrix, given standardisation):

$$\Sigma = \frac{1}{T-1} X^{\top} X$$

We then perform an decomposition of the covariance matrix to solve for the eigenvalues λ and eigenvectors v: $\Sigma v = \lambda v$.

The eigenvectors, sorted by the magnitude of their corresponding eigenvalues, represent the Principal Components or "loadings." These vectors define the directions of maximum variance in the data. The eigenvalues represent the amount of variance explained by each corresponding component. The first principal component (PC1) accounts for the largest possible variance, the second (PC2) accounts for the largest remaining variance while being orthogonal to the first, and so on.

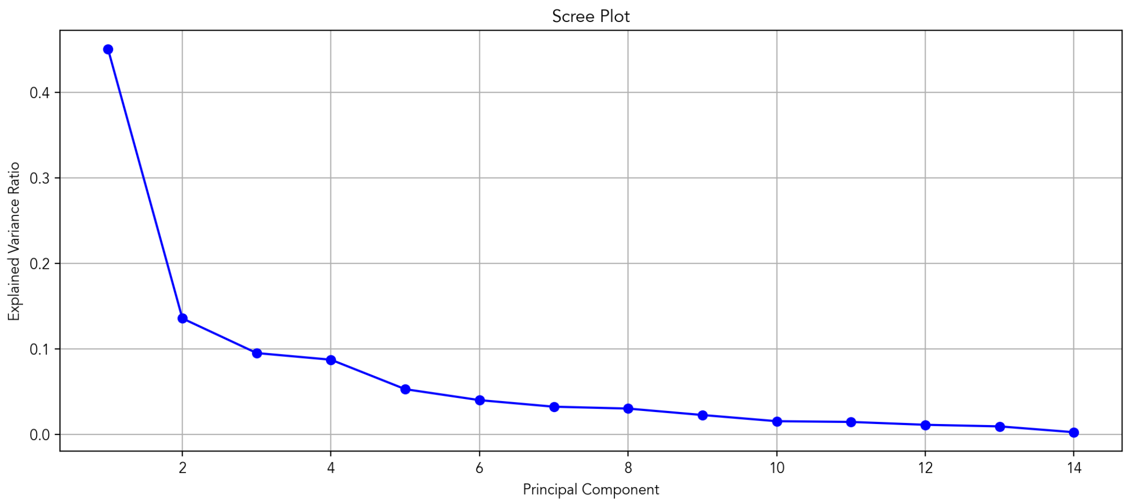

Figure 10. Scree plot for Principal Components.

Source: Own elaboration

3.3.2. Factor identification and interpretation

The selection of the optimal number of factors is guided by the Scree Plot (Figure 10), which displays the proportion of total variance explained by each successive principal component. The plot reveals a dominant first component (PC1) that alone accounts for over 45% of the total variance, reflecting the high degree of correlation across global financial markets. The "elbow" of the curve—the point where the marginal gain in explained variance diminishes significantly—appears after the fifth component. Consequently, we retain the first five Principal Components (PC1–PC5) for analysis, as they collectively capture the vast majority of the systematic risk structure while filtering out idiosyncratic noise.

To interpret the financial meaning of these latent factors, we examine the Principal Component Loadings. This chart illustrates the correlation between each original asset class and the new orthogonal components. By analysing the magnitude and sign of the loadings (the height of the bars), we assign the following distinct economic profiles to the five retained components:

- **PC1 (Global Market Beta):** Characterised by uniformly positive loadings across almost all risky assets—particularly equities (S&P 500, Euro Stoxx 50) and High Yield credit—this factor represents the broad "risk-on" market sentiment. It drives the synchronous movement of global capital markets.

- **PC2 (Credit/Duration):** This factor captures the classic "flight to safety" dynamic. It exhibits strong positive loadings on defensive assets like Investment Grade bonds (US IG, EU IG) and Gold, contrasted against negative loadings on major equity indices, representing the trade-off between safety/duration and equity risk.

- **PC3 (International vs US):** A geographic differentiator that separates the performance of US indices (negative loadings on Nasdaq 100/S&P 500) from International and Emerging Markets (positive loadings on Japan, MSCI EM).

- **PC4 (Real Assets/Inflation):** This factor is distinct and specialised, driven almost exclusively by very high positive loadings on Commodities and Gold, serving as a proxy for inflation hedging or raw material cycles.

- **PC5 (Growth vs. Value):** A style-based factor that contrasts high-growth, tech-heavy markets (positive loadings on Nasdaq 100 and Japan) against traditional, value-heavy European indices (negative loadings on UK FTSE, Euro Stoxx 50).

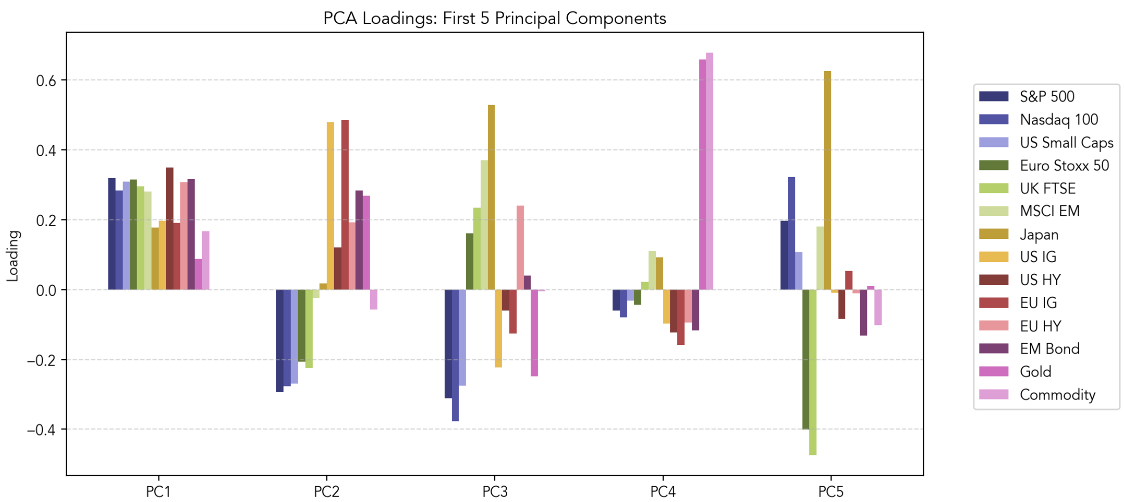

Figure 11. PC loadings per asset class.

Source: Own elaboration

3.3.3. ETF factor exposure

With the economic identity of the five Principal Components established, the final analytical step is to determine the specific sensitivity of each ETF to these orthogonal risk drivers. Unlike the initial asset-class regression, which suffered from multicollinearity, projecting ETF returns onto the uncorrelated Principal Components allows for a robust and stable attribution of risk.We achieve this by treating the derived Principal Components as independent risk factors in a multi-factor model. Formally, for each ETF i, we estimate the following time-series regression equation:

$$R_{i,t} = \alpha_i + \beta_{i,1}PC_{1,t} + \beta_{i,2}PC_{2,t} + \beta_{i,3}PC_{3,t} + \beta_{i,4}PC_{4,t} + \beta_{i,5}PC_{5,t} + \epsilon_{i,t}$$

where:

- $R_{i,t}$ is the standardised daily log return of i-th ETF at time t;
- $PC_{k,t}$ is the return of the k-th Principal Component (factor) at time t;
- α_i represents the unexplained intercept (or idiosyncratic drift);
- $\beta_{i,k}$ is the PCA Beta or factor loading, measuring the sensitivity of the ETF to the k-th factor.

The resulting PCA Betas provide a precise "genetic code" for each fund. A high positive $\beta_{i,1}$ indicates a fund that rallies strongly with global risk sentiment, while a significant negative $\beta_{i,5}$ would identify a fund inversely correlated with the Growth/Tech factor (i.e., a value-focused fund).

Figure 13 displays the PCA-based attribution of total ETF returns, using factor betas to calculate cumulative contributions. It is evident that the top-performing ETFs benefited from a potent combination of broad market exposure (PC1) fused with a US-bias (PC3) and a strong Growth/Tech focus (positive PC5), while maintaining minimal hedging via bond exposure. The middle segment of the distribution, comprising average-performing and generally lower-risk funds, is characterised by a pronounced negative PC5 contribution, signalling a distinct Value focus. Finally, the weakest ETFs underperformed mainly due to high fixed income exposure, indicated by large negative contributions from the Credit/Duration factor (PC2).

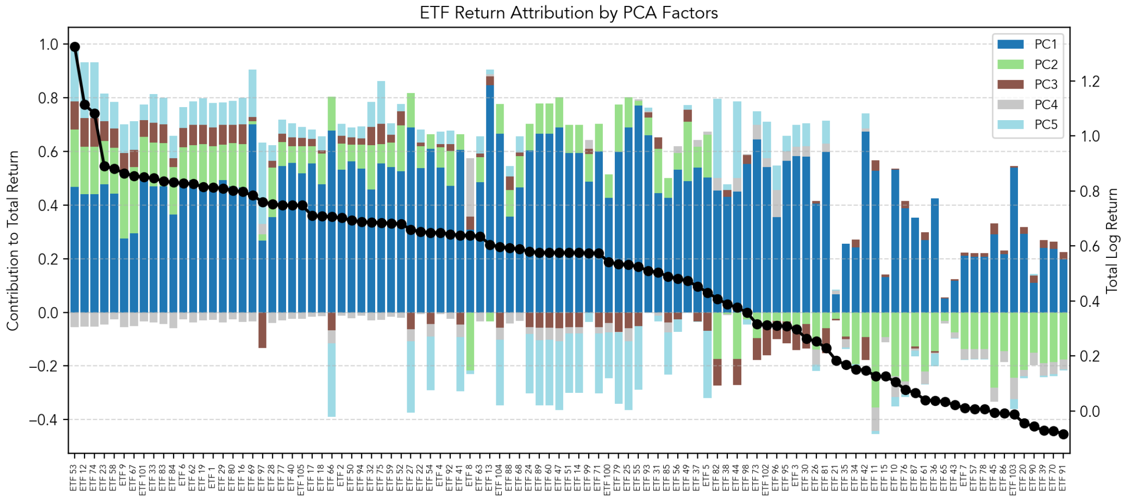

Figure 12. PCA-based contributions to the total return.

Source: Own elaboration

3.4. Combined results and classification

The combined analysis using both OLS regression and PCA factor attribution unequivocally identifies structural asset allocation.To pinpoint the structural drivers of performance, we synthesise findings from both OLS regression and PCA factor attribution into a unified forensic analysis. Figure 14 visualises this comparison for the top, median, and bottom five ETFs, revealing the distinct asset allocation profiles that defined each performance tier.

The Top Performing ETFs (e.g., ETF 53, ETF 12) were driven by a concentrated "pure play" exposure to aggressive US equity risk. OLS results show near-unitary sensitivity to the Nasdaq 100 and S&P 500, with absolutely no defensive hedging. This aggressive positioning is confirmed by PCA, where positive loadings on PC1 (Global Beta) and negative loadings on PC2 (Credit) and PC3 (International) reveal a strategy that systematically shorted safety and diversification. Crucially, their returns were amplified by a high positive loading on PC5, confirming that their outperformance was fueled specifically by the Growth/Tech factor rather than generic market beta.

The Median Performing ETFs (e.g., ETF 99, ETF 14) reveal a structural bias distinct from both the leaders and laggards. Their OLS coefficients shift away from the US, dominated instead by Euro Stoxx 50 and UK FTSE, identifying them as primarily European equity portfolios. PCA mathematically isolates this geographic displacement: these funds combine positive loadings on PC3 (International) with a distinct negative sensitivity to PC5 (Value). Consequently, their "average" results reflect the opportunity cost of holding equities that missed the US tech premium, placing them on the wrong side of the decade’s primary AI-trade.

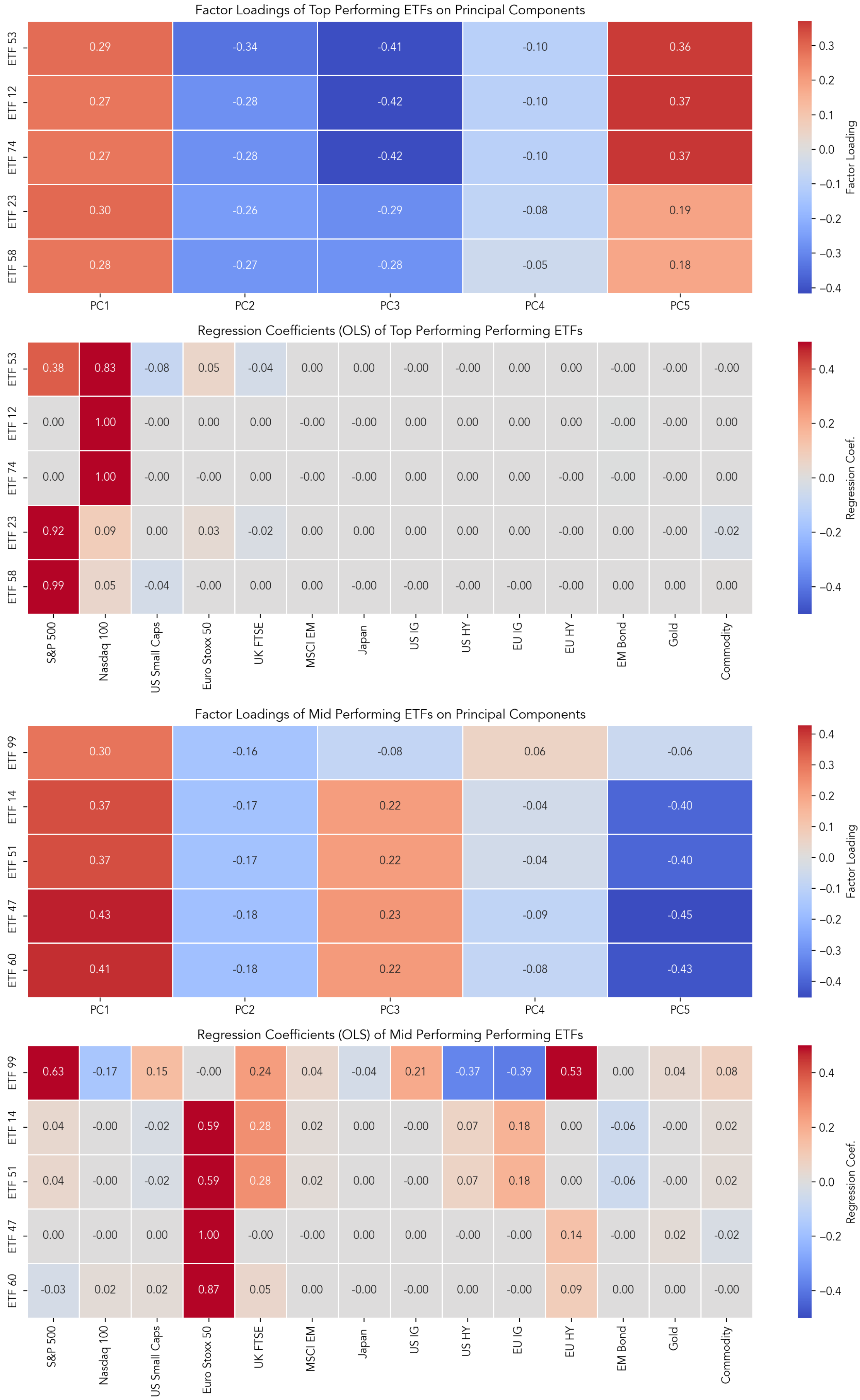

Figure 13. Hybrid identification of 5 Top and Mid performing ETFs.

Source: Own elaboration

The Worst Performing ETFs (e.g., ETF 91, ETF 70) underperformed due to structural asset allocation rather than selection error. OLS analysis exposes a stark "bond-heavy" DNA, with statistically significant loadings on Investment Grade bonds and virtually zero equity exposure. This is robustly corroborated by PCA, where these funds exhibit their highest sensitivity to PC2 (Credit/ Duration)—the defensive "flight-to-safety" factor—while displaying negligible exposure to the high-growth PC5 factor. Ultimately, these portfolios paid the systemic price of being positioned as "safe havens" in a market that overwhelmingly rewarded risk.

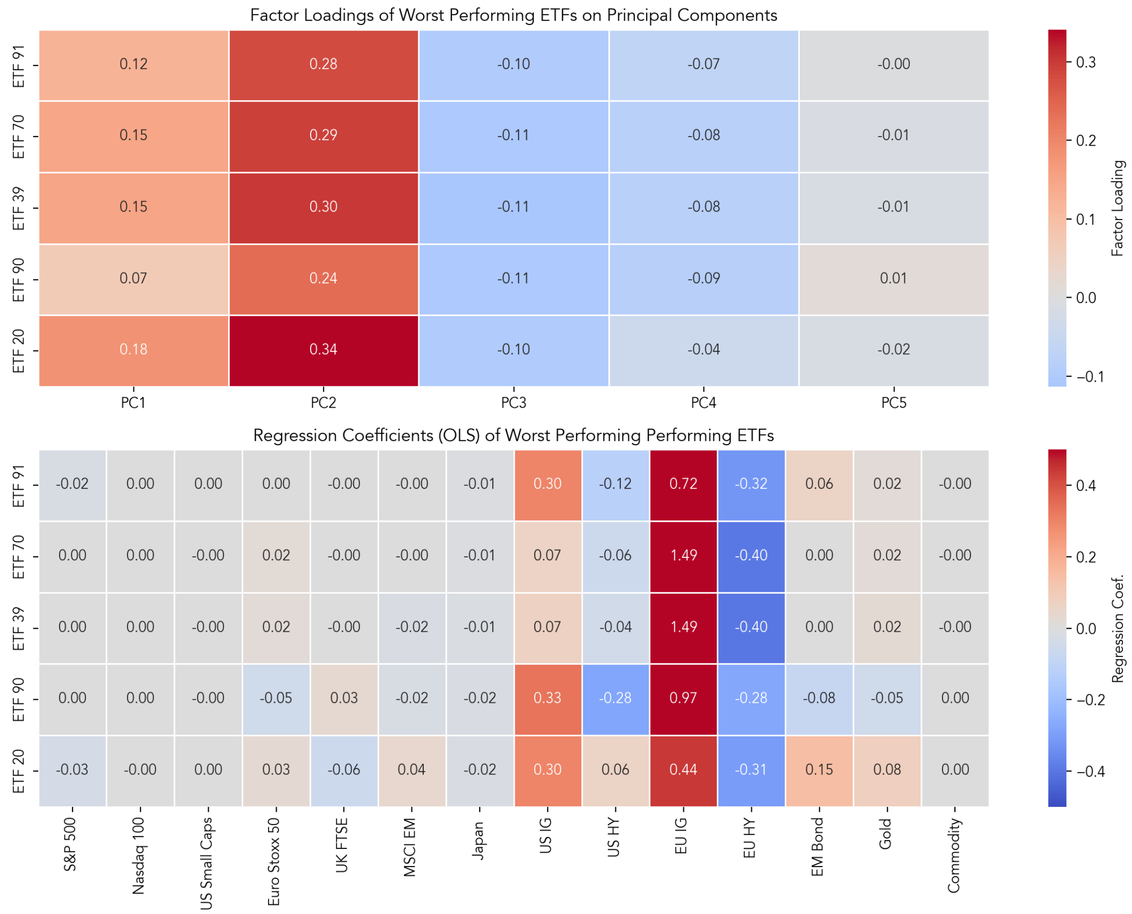

Figure 14. Hybrid identification of 5 Worst performing ETFs.

Source: Own elaboration

4. Mystery allocation

To reverse-engineer the composition of the two unidentified portfolios (MA1 and MA2), we employ Constrained Least Squares Optimisation. This technique solves for the optimal vector of weights that minimises the tracking error (sum of squared return deviations) between the mystery portfolio and our available investment universe, subject to "long-only" and "fully invested" constraints.

Formally, we define the mystery portfolio's return at time t as $R_{M,t}$ and the universe of N =101 available ETFs ($R_{j,t}$). The objective is to find the vector of weights $\omega = [\omega_1, \ldots, \omega_N]^\top$ that minimises the sum of squared residuals over the observation period T, subject to long-only and fully-invested constraints:

$$\begin{aligned} \underset{\omega}{\text{minimize}} \quad & \sum_{t=1}^{T} \left(R_{M,t} - \sum_{j=1}^{N} \omega_j R_{j,t} \right)^2 \\ \text{subject to} \quad & \sum_{j=1}^{N} \omega_j = 1, \\ & \omega_j \geq 0, \quad \forall j = 1, \ldots, N. \end{aligned}$$

4.1. Static portfolio

For the static portfolio, we solve for a single global optimisation problem, with constant set of weights over the entire five-year period. The constrained optimisation for MA1 reveals a highly concentrated portfolio structure, heavily reliant on a few key drivers. The top three positions—ETF 31 (20%), ETF 40 (18%), and ETF 53 (15%)—collectively command over 53%of the total capital. The remaining allocation is distributed across a "long tail" of smaller positions ranging from 10% to 2% (e.g., ETF 8, ETF 11). This distribution suggests a "Core-Satellite" strategy, where the portfolio performance is primarily dictated by three high-conviction bets, supplemented by smaller, likely diversifying or tactical exposures.

Table 4. Optimised weights for MA1 composition (filtered by: 1% <).

	weight (rounded)
ETF 31	0.20
ETF 40	0.18
ETF 53	0.15
ETF 8	0.10
ETF 11	0.10
ETF 10	0.05
ETF 102	0.05
ETF 5	0.05
ETF 34	0.05
ETF 1	0.04
ETF 47	0.02

Source: Own elaboration

4.2. Dynamic portfolio

For the time-varying portfolio, we utilise a Rolling Constrained Optimisation with a 252-day look-back window. This iteratively solves for the optimal weights at each time step based on the most recent year of data. The initial rolling optimisation for the dynamic portfolio (MA2) produced a volatile output, characterised by frequent shifts and significant noise allocations (Figure 15). To isolate the true tactical intent of the MA2, we applied a Regime-Based Filtering Rule. By visually inspecting the allocation timeline, we identified three distinct periods of stability: "Period 1" (Feb 2020 – Oct 2020), "Period 2" (Aug 2021 – Oct 2022), and "Period 3" (Nov 2023 – May 2024). We then filtered the dataset to retain only those ETFs that maintained a dominant average weight (greater than 5%) within at least one of these stable windows. This process successfully revealed a sequence of higher-conviction bets, allowing for a precise reconstruction of the strategy. Figure 16 displays the cumulative performance of the fitted dynamic allocation against the MA2.

Figure 15. Dynamic weights for MA2: all vs filtered.

Source: Own elaboration

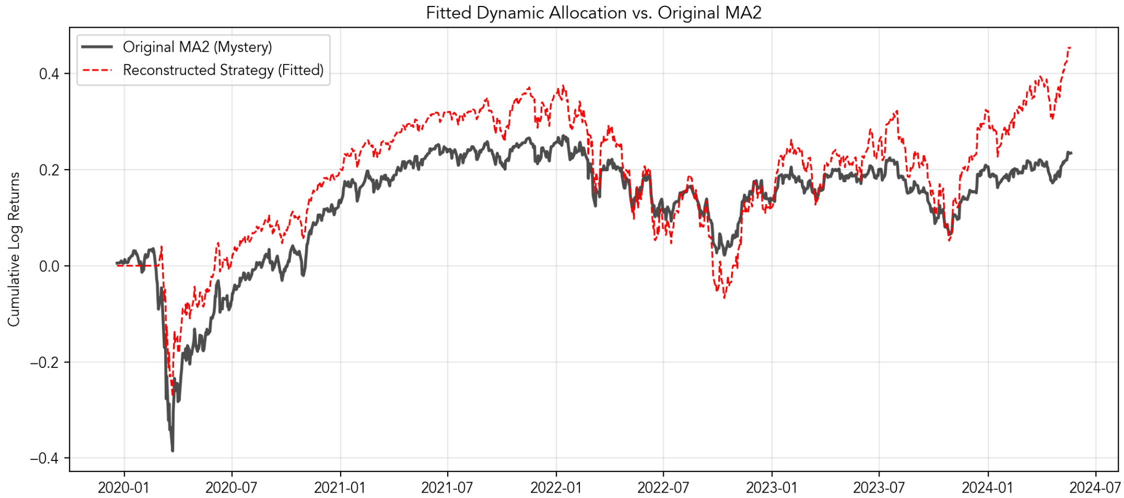

Figure 16. Fitted MA2.

Source: Own elaboration

4.3. Model fit and accuracy

To validate our reconstruction, we calculated key error metrics comparing our replicated portfolios against the original mystery series. As shown in the table below, the Static (MA1) model achieves a near-perfect fit (R2 virtually at 1), confirming that the mystery portfolio is indeed a constant-mix strategy comprised of the identified ETFs.

The Dynamic (MA2) reconstruction, while less precise due to the regime-based simplification, still demonstrates strong explanatory power. A correlation of 0.90 confirms that our "Regime-Switching" model successfully captures the strategy's primary tactical moves, even after filtering out the optimiser's daily noise.

Table 5. Goodness-of-fit metrics for identified allocations.

Model	Correlation	R^2	RMSE (Ann.)	Tracking Error (Ann.)
Static (MA1)	1.000	0.9999	0.0011	0.0010
Dynamic (MA2)	0.896	0.7115	0.0941	0.0941

Source: Own elaboration

4.4. Implied asset allocations

Additionally, we undergo the Mystery Allocations through our hybrid OLS-PCA treatment to identify the asset classes underlying the composition of portfolios (Figure 17). The analysis exposes MA1 as a diversified, constant-mix strategy with stable loadings on both broad equity (S&P 500) and fixed income factors (US and EU IG, Gold), consistent with a balanced "Core" profile. Conversely, MA2 reveals a heavier equity focused, diversified between US and global markets, while avoiding overly growth-oriented stocks. This is further underlined once we run the dynamic optimisation on assets classes instead of ETFs (Figure 18). First period is dominated by S&P 500, with MSCI EM and Euro Stoxx 50 coming right after, second gives priority to Europe and EM equities with no US exposure and third becomes the most diversified, with Japan and Nasdaq taking the lead.

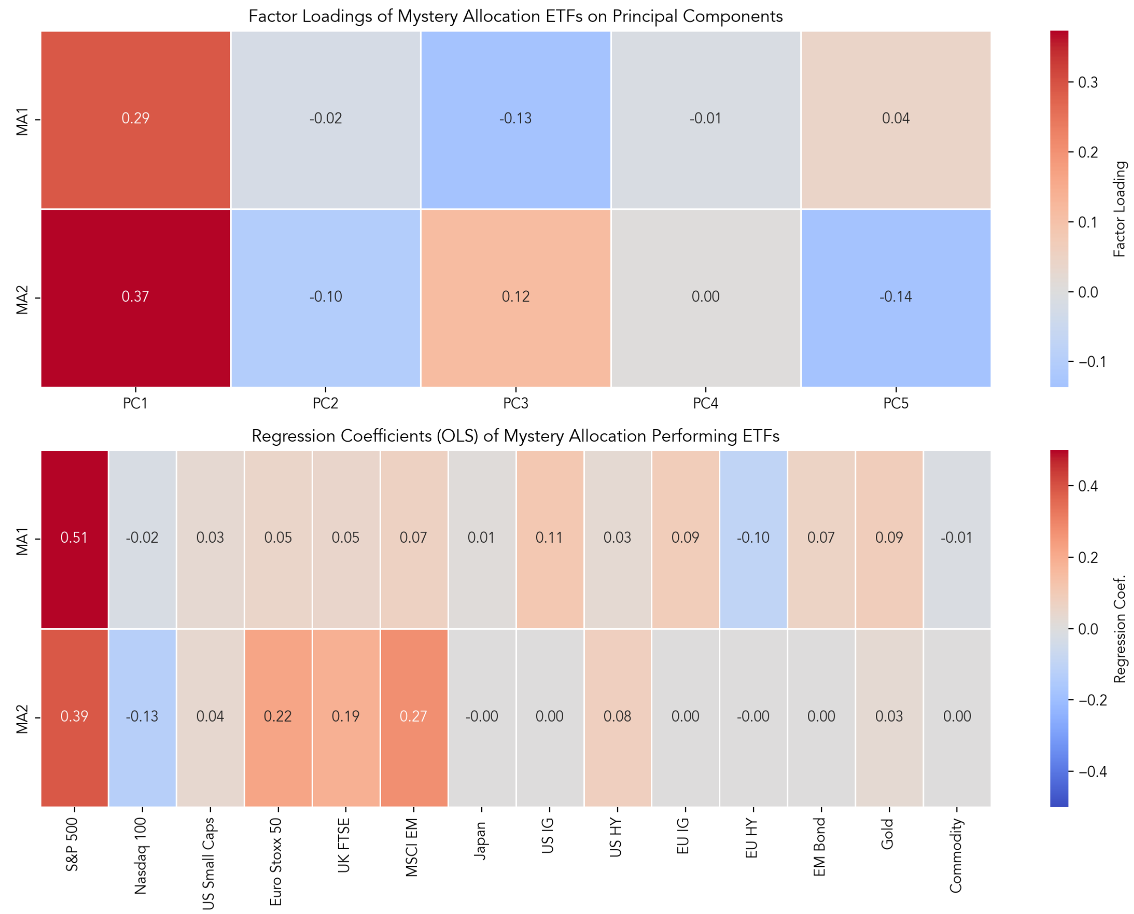

Figure 17. Asset classification for MAs.

Source: Own elaboration

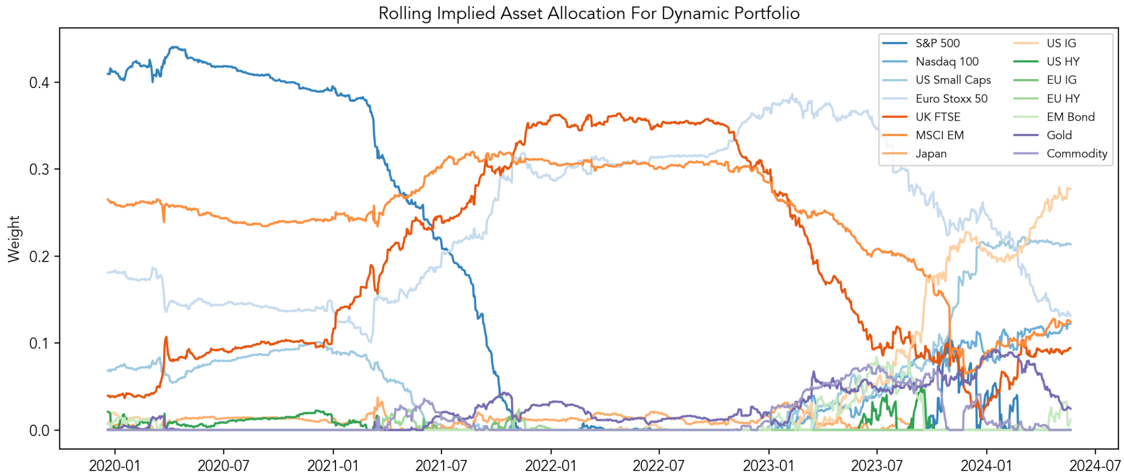

Figure 18. 15. Dynamic weights for MA2 (asset class based).

Source: Own elaboration

CONCLUSION

Our study integrated descriptive statistics, OLS regression, Principal Component Analysis and constrained optimisation to deconstruct the performance drivers of 101 anonymous ETFs and reverse-engineer two mystery portfolios over a volatile five-year period (2020–2024). Initial analysis of core performance metrics confirmed a classical risk-reward tradeoff, revealing a sharp division of the universe into two distinct clusters: low-volatility/low-reward defensive funds and high-volatility/high-return growth strategies.

This performance divergence was dictated primarily by structural asset allocation rather than idiosyncratic selection. The Top Performers delivered exceptional returns by aggressively isolating US Growth and Technology exposure, effectively riding the post-pandemic tech rally. In contrast, the Bottom Performers suffered a "safety penalty," heavily burdened by exposure to safe-haven assets like bonds and commodities, which acted as a drag during the inflationary regime.

The analytical framework culminated in the successful identification of the two mystery strategies. Mystery Allocation 1 (MA1) was revealed as a disciplined "Core-Satellite" approach that balanced permanent growth kickers (US, often tech equities) with investment-grade US/EU bonds. Mystery Allocation 2 (MA2) was identified as a geographically rotating, equity-focused portfolio, which pivoted from a defensive, value-oriented stance during the 2020–2022 volatility to a concentrated "Risk-On" bet on US Tech in late 2023.